

$y'' + p(z)y' + q(z)y = 0$, želimo rešitev v ok. z_0 .

1) p, q holomorfni v ok. $z_0 \Rightarrow y$ holomorfna, $y = \sum_{n=0}^{\infty} c_n (z-z_0)^n$ enolično določena z pogoji $y(z_0) = \dots$
 $y'(z_0) = \dots$

2) $p(z)$ ima v z_0 pol st. najv. 1, $q(z) - 11$ najv. 2

$$p_0 = \lim_{z \rightarrow z_0} (z-z_0)p(z) \quad \left\{ \begin{array}{l} \mu(\mu-1) + p_0\mu + q_0 = 0 \text{ (ničli } \mu_1, \mu_2) \\ a) \mu_1 - \mu_2 \notin \mathbb{Z} \end{array} \right.$$

$$q_0 = \lim_{z \rightarrow z_0} (z-z_0)^2 q(z)$$

spl. reš.:

$$y_1 = (z-z_0)^{\mu_1} \sum_{n=0}^{\infty} c_n (z-z_0)^n$$

$$y_2 = (z-z_0)^{\mu_2} \sum_{n=0}^{\infty} d_n (z-z_0)^n$$

b) $\operatorname{Re}(\mu_1) \geq \operatorname{Re}(\mu_2)$, $\mu_1 - \mu_2 \in \mathbb{N} \cup \{0\}$

$$y_1 = (z-z_0)^{\mu_1} \sum_{n=0}^{\infty} c_n (z-z_0)^n \quad \text{vzamemo večjo}$$

y_2 dobimo s pomočjo $\det W$:

$$y_2 = C y_1 \ln(z-z_0) + (z-z_0)^{\mu_2} \sum_{n=0}^{\infty} d_n (z-z_0)^n$$

① $(1+z^2)y'' - 4zy' + 6y = 0$, $z_0 = 0$

$$\rightarrow y'' - \underbrace{\frac{4z}{1+z^2}}_{p(z)} y' + \underbrace{\frac{6}{1+z^2}}_{q(z)} y = 0$$

p, q holomorfni ok. 0

$$y = \sum_{n=0}^{\infty} c_n z^n$$

$$y' = \sum_{n=1}^{\infty} c_n n z^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) z^{n-2}$$

$$\sum_{n=2}^{\infty} n(n-1) c_n z^{n-2} + \sum_{n=2}^{\infty} n(n-1) c_n z^n - 4 \sum_{n=1}^{\infty} n c_n z^n + 6 \sum_{n=0}^{\infty} c_n z^n = 0$$

$$\downarrow m=n-2$$

$$\sum_{m=0}^{\infty} (m+2)(m+1) c_{m+2} z^m$$

$\rightarrow b_n$: (glejamo koef. pri z^n)

$$n=0: 2c_2 + 6c_0 = 0 \rightarrow c_2 = -3c_0$$

$$n=1: 6c_3 - 4c_1 + 6c_1 = 0 \rightarrow c_3 = -\frac{1}{3}c_1$$

$$n \geq 2: (n+2)(n+1)c_{n+2} + n(n-1)c_n - 4nc_n + 6c_n = 0$$

$$c_{n+2} = -\frac{n(n-1)-4n+6}{(n+2)(n+1)} c_n = -\frac{n^2-5n+6}{(n+2)(n+1)} c_n = -\frac{(n-2)(n-3)}{(n+2)(n+1)} c_n$$

c_0	$\mathbb{C}$
c_1	$\mathbb{C}$
c_2	$-3c_0$
c_3	$-\frac{1}{3}c_1$
c_4	0
c_5	0
$\vdots$	$\vdots$

$$\rightarrow y = c_0 + c_1 z - 3c_0 z^2 - \frac{1}{3}c_1 z^3 = \boxed{c_0(1-3z^2) + c_1(z - \frac{z^3}{3})} \text{ spl. reš.}$$

$$(2.) y'' = zy, z_0 = 0$$

$$\left. \begin{aligned} y &= \sum_{n=0}^{\infty} c_n z^n \\ y' &= \sum_{n=1}^{\infty} n c_n z^{n-1} \\ y'' &= \sum_{n=2}^{\infty} n(n-1) c_n z^{n-2} \end{aligned} \right\} \begin{aligned} \sum_{n=2}^{\infty} n(n-1) c_n z^{n-2} &= \sum_{n=0}^{\infty} c_n z^{n+1} \\ \sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} z^n &= \sum_{n=1}^{\infty} c_{n-1} z^n \end{aligned}$$

$$n=0: 2c_2 = 0$$

$$n \geq 1: (n+2)(n+1)c_{n+2} = c_{n-1} \rightarrow c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$$

$$\rightarrow y = c_0 + c_1 z + \frac{c_0}{2 \cdot 3} z^3 + \frac{c_1}{3 \cdot 4} z^4 + \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6} z^6 + \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7} z^7 + \dots$$

$$= c_0 \left(1 + \frac{z^3}{2 \cdot 3} + \frac{z^6}{2 \cdot 3 \cdot 5 \cdot 6} + \dots \right) + c_1 \left(z + \frac{z^4}{3 \cdot 4} + \frac{z^7}{3 \cdot 4 \cdot 6 \cdot 7} + \dots \right)$$

c_0	c
c_1	c
c_2	0
c_3	$\frac{c_0}{2 \cdot 3}$
c_4	$\frac{c_1}{4 \cdot 3}$
c_5	0
c_6	$\frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6}$
c_7	$\frac{c_1}{4 \cdot 3 \cdot 7 \cdot 6}$
c_8	0

$$(3.) zzy'' + (1 - 2z^2)y' - 4zy = 0, z_0 = 0$$

$$y'' + \frac{1-2z^2}{2z} y' - 2y = 0$$

$$\left. \begin{aligned} p_0 &= \lim_{z \rightarrow 0} \frac{1-2z^2}{2z} = \frac{1}{2} \\ q_0 &= \lim_{z \rightarrow 0} (-2z^2) = 0 \end{aligned} \right\} \begin{aligned} \mu(\mu-1) + \frac{1}{2}\mu &= 0 \\ \mu^2 - \frac{1}{2}\mu &= 0 \\ \mu(\mu - \frac{1}{2}) &= 0 \end{aligned} \quad \begin{aligned} \mu_1 &= 0 \\ \mu_2 &= \frac{1}{2} \end{aligned}$$

$$\rightarrow y_1 = \sum_{n=0}^{\infty} c_n z^n \rightarrow 2 \sum_{n=2}^{\infty} n(n-1) c_n z^{n-1} + \sum_{n=1}^{\infty} n c_n z^{n-1} - 2 \sum_{n=1}^{\infty} n c_n z^{n+1} - 4 \sum_{n=0}^{\infty} c_n z^{n+1} = 0$$

$$2 \sum_{n=1}^{\infty} (n+1)(n) c_{n+1} z^n + \sum_{n=0}^{\infty} (n+1) c_{n+1} z^n - 2 \sum_{n=2}^{\infty} (n-1) c_{n-1} z^n - 4 \sum_{n=1}^{\infty} c_{n-1} z^n = 0$$

$$n=0: c_1 = 0$$

$$n=1: c_2 = \frac{2}{3} c_0$$

$$n \geq 2: n(n+1)c_{n+1} + (n+1)c_{n+1} - 2(n-1)c_{n-1} - 4c_{n-1} = 0 \quad \left\{ \begin{aligned} y_1 &= c_0 \left(1 + \frac{2}{3} z^2 + \frac{2 \cdot 2}{3 \cdot 7} z^4 + \frac{2 \cdot 2 \cdot 2}{3 \cdot 7 \cdot 11} z^6 + \dots \right) \end{aligned} \right.$$

$$(n+1)^2 c_{n+1} - (2n+2)c_{n-1} = 0 \rightarrow c_{n+2} = \frac{2c_n}{2n+3}$$

$$y_2 = z^{\frac{1}{2}} \sum c_n z^n = \sum_{n=0}^{\infty} c_n z^{n+\frac{1}{2}}$$

$$y_2 = c_0 \sqrt{z} \left(1 + \frac{z^2}{2} + \frac{z^4}{2 \cdot 4} + \frac{z^6}{2 \cdot 4 \cdot 6} + \dots \right) = c_0 \sqrt{z} \sum_{n=0}^{\infty} \frac{z^{2n}}{2^n n!} = \underline{\underline{c_0 \sqrt{z} \exp\left(\frac{z^2}{2}\right)}}$$

$$(4) z_0 = 0, z^2 y'' + z(z+1)y' - y = 0$$

$$y'' = \frac{(z+1)}{z} y' - \frac{1}{z^2} y = 0$$

$$\left. \begin{array}{l} p_0 = 1 \\ q_0 = -1 \end{array} \right\} \begin{array}{l} \mu(\mu-1) + \mu - 1 = 0 \\ \mu^2 - 1 = 0 \end{array}$$

$$\begin{array}{l} \mu_1 = 1 \\ \mu_2 = -1 \end{array}$$

$$\mu_2 = -1:$$

$$y = z^{-1} \sum_{n=0}^{\infty} c_n z^n \rightarrow \sum_{n=0}^{\infty} c_n (n-1)(n-2) z^{n-1} + \sum_{n=0}^{\infty} c_n (n-1) z^n + \sum_{n=0}^{\infty} c_n (n-1) z^{n-1} - \sum_{n=0}^{\infty} c_n z^{n-1} = 0$$

$$\sum_{n=1}^{\infty} c_{n+1} (n)(n-1) z^n + \sum_{n=0}^{\infty} c_n (n-1) z^n + \sum_{n=1}^{\infty} c_{n+1} n z^n - \sum_{n=1}^{\infty} c_{n+1} z^n = 0$$

$$n = -1:$$

$$c_0 (-1)(-2) + c_0 (-1) - c_0 = 0 \rightarrow c_0 \in \mathbb{C}$$

$$n > -1:$$

$$c_{n+1} (n)(n-1) + c_n (n-1) + c_{n+1} n - c_{n+1} = 0$$

$$c_{n+1} = -\frac{n-1}{n^2 - n + n - 1} c_n = -\frac{c_n}{n+1} \quad (n \neq 1)$$

$$n = 1:$$

$$c_2 (1)(0) + c_1 \cdot 0 + c_2 \cdot 1 - c_2 = 0 \rightarrow c_2 \in \mathbb{C}$$

$$\begin{array}{c|l} c_0 & \mathbb{C} \\ c_1 & -c_0 \\ c_2 & \mathbb{C} \\ c_3 & -\frac{c_2}{3} \\ c_4 & -\frac{c_3}{4} = \frac{c_2}{12} \\ c_5 & -\frac{c_2}{3 \cdot 4 \cdot 5} \\ c_i & (-1)^{i+1} \frac{c_2 2^i}{i!} \quad i \geq 3 \end{array}$$

$$\rightarrow y_1 = \frac{1}{z} \left(c_0 - c_0 z + c_2 \left(z^2 - \frac{1}{3} z^3 + \frac{1}{3 \cdot 4} z^4 - \dots \right) \right) =$$

$$= c_0 \underbrace{\frac{1-z}{z}}_{u_1(z)} + c_2 \underbrace{\left(z - \frac{1}{3} z^2 + \frac{1}{3 \cdot 4} z^3 - \dots \right)}_{u_2(z)} = \frac{1-z}{z} c_0 + \left(\frac{2e^{-z} - 1 + z}{z} \right) c_2$$

• Je u_1, u_2 lin. neodv.: je to ze resitev

$$(5) 4z^2 y'' - 4z y' + (1-2z)y = 0, z_0 = 0$$

$$p = -1, q = \frac{1-2z}{4z^2}$$

$$p_0 = 0, q_0 = \frac{1}{4}$$

$$\mu(\mu-1) + \frac{1}{4} = 0$$

$$\mu^2 - \mu + \frac{1}{4} = 0$$

$$\left(\mu - \frac{1}{2} \right)^2 = 0$$

$$\mu_{1,2} = \frac{1}{2}$$

$$y = z^{\frac{1}{2}} \sum c_n z^n = \sum_{n=0}^{\infty} c_n z^{n+\frac{1}{2}} \Rightarrow \dots \rightarrow c_0 \in \mathbb{C}$$

$$c_n = \frac{1}{n} c_{n-1} = \frac{1}{n!} c_0$$

$$\rightarrow y_1 = z^{\frac{1}{2}} \sum_{n=0}^{\infty} \frac{c_0}{n!} z^n = z^{\frac{1}{2}} e^z$$

$$y_2 = y_1 \ln z + \sum_{n=0}^{\infty} c_n z^{n+\frac{1}{2}}$$

$$y_2' = y_1' \ln z + \frac{y_1}{z} + \sum_{n=0}^{\infty} c_n (n+\frac{1}{2}) z^{n-\frac{1}{2}}$$

$$y_2'' = y_1'' \ln z + \frac{y_1'}{z} + \frac{y_1'}{z} - \frac{y_1}{z^2} + \sum_{n=0}^{\infty} c_n (n+\frac{1}{2})(n-\frac{1}{2}) z^{n-\frac{3}{2}}$$

$$\rightarrow \underbrace{\ln(z^2 y'' - 4z^2 y' + (1-2z)y')} + 4z^2 \left(\frac{2y_1'}{z} - \frac{y_1}{z} + \sum_{n=0}^{\infty} c_n (n+\frac{1}{2})(n-\frac{1}{2}) z^{n-\frac{3}{2}} \right) \\ - 4z^2 \left(\frac{y_1}{z} + \sum_{n=0}^{\infty} c_n (n+\frac{1}{2}) z^{n-\frac{1}{2}} \right) + (1-2z) \sum_{n=0}^{\infty} c_n z^{n+\frac{1}{2}} = 0 \\ 4z^2 \left(2 \sum_{n=1}^{\infty} \frac{n+\frac{1}{2}}{n!} z^{n-\frac{3}{2}} - \sum_{n=1}^{\infty} \frac{z^{n-\frac{3}{2}}}{n!} + \sum_{n=0}^{\infty} c_n (n+\frac{1}{2})(n-\frac{1}{2}) z^{n-\frac{3}{2}} \right)$$

$$- 4z^2 \left(\sum_{n=0}^{\infty} \frac{z^{n-\frac{1}{2}}}{n!} + \sum_{n=0}^{\infty} c_n (n+\frac{1}{2}) z^{n-\frac{1}{2}} \right) + \sum_{n=0}^{\infty} c_n z^{n+\frac{1}{2}} - 2 \sum_{n=0}^{\infty} c_n z^{n+\frac{3}{2}} = 0$$

$$z^{\frac{1}{2}}: 8 \cdot \frac{1}{2} - 4 + 4c_0 \left(-\frac{1}{4} \right) + c_0 = 0 \Rightarrow c_0 \in \mathbb{R}$$

$$z^{\frac{3}{2}}: 8 \cdot \frac{3}{2} - 4 + 4c_1 \cdot \frac{3}{2} - 4 - 4c_0 \cdot \frac{1}{2} + c_1 - 2c_0 = 0 \Rightarrow 4 + 4c_1 - 4c_0 = 0 \rightarrow c_1 = c_0 - 1$$

$c_2, \dots$ podobno

Besselove funkcije

$$z^2 y'' + zy' + (z^2 + \nu^2)y = 0 \quad \text{Besselova DE}$$

$$\rightarrow J_{\nu}(z) = \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{z}{2}\right)^{2n+\nu}}{n! \Gamma(n+\nu+1)} \quad (\nu \in \mathbb{R}) \quad \text{Reši Besselovo DE}$$

Lastnosti:

$$1) \nu \notin \mathbb{Z} \Rightarrow J_{\nu}, J_{-\nu} \text{ lin. neodv.}$$

$$\nu \in \mathbb{Z} \Rightarrow J_{-\nu} = (-1)^{\nu} J_{\nu}$$

$$2) \text{ Von Neumanova funkc. (tudi reš. Bes. DE)}$$

$$\nu \notin \mathbb{Z}: Y_{\nu}(z) = \frac{J_{\nu}(z) \cos(\nu\pi) - J_{-\nu}(z)}{\sin(\nu\pi)}$$

$$\nu = n \in \mathbb{N}: Y_{\nu}(z) = \lim_{\nu \rightarrow n} Y_{\nu}(z)$$

$$3) \sum_{n=-\infty}^{\infty} J_n(z) t^n = e^{\frac{z}{2}(t - \frac{1}{t})}$$

$$4) 2J'_{\nu}(z) = J_{\nu-1}(z) - J_{\nu+1}(z)$$

$$5) \frac{2\nu}{z} J_{\nu}(z) = J_{\nu-1}(z) + J_{\nu+1}(z)$$

$$\textcircled{1} J_{\frac{1}{2}}(z) = \sqrt{\frac{2z}{\pi}} \frac{\sin z}{z}. \text{ Pokaži, da za } n=0,1,\dots \text{ velja } J_{n+\frac{1}{2}} = (-1)^n z^n \sqrt{\frac{2z}{\pi}} \left(\frac{1}{z} \frac{d}{dz} \right)^n \left(\frac{\sin z}{z} \right).$$

$$\text{Dobro } J_{-\frac{1}{2}}(z).$$

$$n=0: J_{\frac{1}{2}} = \sqrt{\frac{2z}{\pi}} \frac{\sin z}{z} \checkmark \rightarrow \text{indukcija: ali velja za } n+1?$$

$$J_{n+\frac{1}{2}} \stackrel{?}{=} (-1)^{n+1} z^{n+1} \sqrt{\frac{2z}{\pi}} \left(\frac{1}{z} \frac{d}{dz} \right)^{n+1} \left(\frac{\sin z}{z} \right) = (-1)^{n+1} z^{n+1} \sqrt{\frac{2z}{\pi}} \frac{1}{z} \frac{d}{dz} \left[\left(\frac{1}{z} \frac{d}{dz} \right)^n \left(\frac{\sin z}{z} \right) \right] =$$

$$= (-1)^{n+1} z^n \sqrt{\frac{2z}{\pi}} \frac{d}{dz} \left[\frac{J_{n+\frac{1}{2}}}{(-1)^n z^n \sqrt{\frac{2z}{\pi}}} \right] = -z^n \sqrt{\frac{2z}{\pi}} \frac{d}{dz} \left[\frac{J_{n+\frac{1}{2}}}{z^{n+\frac{1}{2}}} \right] = -z^{n+\frac{1}{2}} \left[J'_{n+\frac{1}{2}} z^{-n-\frac{1}{2}} + J_{n+\frac{1}{2}} \left(-n-\frac{1}{2} \right) z^{-n-\frac{3}{2}} \right]$$

$$= -z^{n+\frac{1}{2}} \left[\frac{1}{2} (J_{n-\frac{1}{2}} - J_{n+\frac{3}{2}}) z^{-n-\frac{1}{2}} - J_{n+\frac{1}{2}} \left(n + \frac{1}{2} \right) z^{-n+\frac{3}{2}} \right] =$$

$$\hookrightarrow \frac{(n+\frac{1}{2}) J_{n+\frac{1}{2}}}{z}$$

$$= - \left[\frac{1}{2} (J_{n-\frac{1}{2}} - J_{n+\frac{3}{2}}) - \frac{1}{2} (J_{n-\frac{1}{2}} + J_{n+\frac{3}{2}}) \right] = J_{n+\frac{3}{2}} \checkmark$$

$$J_{-\frac{1}{2}}(z) = ?$$

$$\frac{z^{\frac{1}{2}}}{z} J_{\frac{1}{2}} = J_{-\frac{1}{2}} + J_{\frac{3}{2}} \rightarrow J_{-\frac{1}{2}} = \frac{1}{z} J'_{\frac{1}{2}} - J_{\frac{3}{2}} = \dots = \underline{\underline{\sqrt{\frac{2z}{\pi}} \frac{\cos z}{z}}}$$

② $J_\nu(x)$, $x \in \mathbb{R}$, $a, b \geq 0$, razl. ničli funkcije $J_\nu(x)$. Pokaži, da je $\int_0^1 x J_\nu(ax) J_\nu(bx) dx = 0$.

$$u(x) = J_\nu(ax) \rightarrow (ax)^2 J''_\nu(ax) + ax J'_\nu(ax) + ((ax)^2 - \nu^2) J_\nu(ax) = 0$$

$$v(x) = J_\nu(bx) \quad u'(x) = a J'_\nu(ax), \quad u''(x) = a^2 J''_\nu(ax)$$

$$\rightarrow \left. \begin{aligned} x^2 u'' + x u' + (a^2 x^2 - \nu^2) u &= 0 \quad / \cdot v \\ x^2 v'' + x v' + (b^2 x^2 - \nu^2) v &= 0 \quad / \cdot u \end{aligned} \right\}$$

$$x^2 u'' v + x u' v + (a^2 x^2 - \nu^2) u v - x^2 v'' u - x v' u - (b^2 x^2 - \nu^2) u v = 0$$

$$\underbrace{x(u''v - uv'') + (u'v - uv')}_{(x(u'v - uv'))'} + uvx(a^2 - b^2) = 0 \quad / \int_0^1 dx$$

$$\int_0^1 x u v dx = \frac{-1}{a^2 - b^2} x(u'v - uv') \Big|_0^1 = -\frac{1}{a^2 - b^2} \left(a \overset{0}{J'_\nu(a)} \overset{0}{J_\nu(b)} - b \overset{0}{J'_\nu(b)} \overset{0}{J_\nu(a)} \right) = 0 \quad \checkmark$$

Opomba: če imamo vekt. pr. $C^0([0,1])$ s skal. pr. $\langle f, g \rangle = \int x f g dx$, so Besselove funkc.

$J_\nu(a_0 x), J_\nu(a_1 x), \dots$ ortogonalne.

($a_0, a_1, \dots$ ničle J_ν)

Legendrovi polinomi

$$P_n(z) = \frac{1}{n!} 2^n \frac{d^n}{dz^n} ((z^2-1)^n) \quad \text{rešija DE } (z^2-1)y'' - 2zy' - n(n+1)y = 0$$

Lastnosti:

$$1) P_n(1) = (-1)^n$$

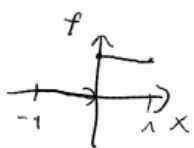
$$2) |P_n(x)| \leq 1; x \in [-1, 1]$$

$$3) (n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

$$4) \int_{-1}^1 P_n(x)P_m(x)dx = \begin{cases} 0; & n \neq m \\ \frac{2}{2n+1}; & n=m \end{cases}$$

$$5) \frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n$$

$$1) f: [-1, 1] \rightarrow \mathbb{R}, f(x) = \begin{cases} 0; & x \in [-1, 0) \\ 1; & x \in [0, 1] \end{cases} \text{ Razvij } f \text{ po Leg. pol.}$$



$$f(x) = a_0 P_0 + a_1 P_1 + \dots; a_n = ?$$

$$a_n = \frac{\langle f, P_n \rangle}{\langle P_n, P_n \rangle} = \frac{2n+1}{2} \int_{-1}^1 f(x) P_n(x) dx = \frac{2n+1}{2} \int_0^1 P_n(x) dx$$

$$P_n(x) = \frac{1}{n!} 2^n \underbrace{((x^2-1)^n)^{(n)}}_{\text{soda } \forall n}$$

a) n sod: P_n soda (sledi iz odred sode f. je liha)

$$\rightarrow a_n = \frac{2n+1}{2} \frac{1}{2} \int_{-1}^1 P_n(x) P_0(x) dx = \begin{cases} 0; & n=2, 4, \dots \\ \frac{1}{2}; & n=0 \end{cases}$$

b) n liha: P_n liha

$$\sum_{n=0}^{\infty} P_n(x)t^n = \frac{1}{\sqrt{1-2xt+t^2}} \Big|_0^1 dx$$

predt, da je $|t| < 1$

$$\sum_{n=0}^{\infty} t^n \int_0^1 P_n(x) dx = \int_0^1 \frac{1}{\sqrt{1-2xt+t^2}} dx = \dots = -\frac{1}{t} (\sqrt{(1-t)^2} - \sqrt{1+t^2}) = -\frac{1}{t} (1-t - \sqrt{1+t^2}) =$$

$$= -\frac{1}{t} + 1 + \frac{\sqrt{1+t^2}}{t} = 1 - \frac{1}{t} + \frac{1}{t} \sum_{n=0}^{\infty} \binom{1/2}{n} t^{2n} = 1 - \cancel{\frac{1}{t}} + \frac{1}{t} \left(1 + \binom{1/2}{1} t^2 + \dots \right) =$$

$$= 1 + \sum_{n=1}^{\infty} \binom{1/2}{n} t^{2n-1} \Rightarrow \binom{1/2}{n} = \int_0^1 P_{2n-1} dx$$

$$\int_0^1 P_1 dx = \frac{1}{2}$$

$$\int_0^1 P_3 dx = -\frac{1}{8}$$

;

Parcialne dif. en. in metoda ločitve spr.

$$u(x,t) = X(x) \cdot T(t) \rightarrow \text{PDE} \rightarrow \dots \rightarrow X(x)=?, T(t)=?$$

1) Poišči $u: [0, \pi] \times [0, \infty) \rightarrow \mathbb{R}$, ki zadošča $u_t = u_{xx}$; $u(x, 0) = \begin{cases} 0; & x \in [0, \pi/2) \\ 1; & x \in [\pi/2, \pi] \end{cases}$ in $u(0, t) = u(\pi, t) = 0$

$$u = X(x) T(t)$$

$$\rightarrow X \dot{T} = X'' T \quad \frac{\dot{T}}{T} = \frac{X''}{X} = \lambda$$

a) $X'' - \lambda X = 0$

$$\lambda^2 - \lambda = 0$$

$$\lambda_{1,2} = \pm \sqrt{\lambda}$$

i) $\lambda > 0$

$$X = A_1 e^{\sqrt{\lambda} x} + A_2 e^{-\sqrt{\lambda} x}$$

zač. pogoji $A_1 = A_2 = 0$

Dobimo samo trivialno reš. $\rightarrow$ -1-

ii) $\lambda = 0$

$$X = kx + n$$

zač. pogoji $k = n = 0$

iii) $\lambda < 0$

$$X = A_1 e^{i\sqrt{|\lambda|} x} + A_2 e^{-i\sqrt{|\lambda|} x}$$

zapišemo rajč kot

$$X(x) = A_1 \sin \sqrt{|\lambda|} x + A_2 \cos \sqrt{|\lambda|} x$$

$\rightarrow$ zač. pog. $\rightarrow X(\pi) = A_1 \sin \sqrt{|\lambda|} \pi = 0$

$$\rightarrow \lambda = -n^2, \quad \underline{X_n(x) = \sin(nx)}$$

$$\Rightarrow \frac{\dot{T}}{T} = -n^2 \rightarrow T_n = B_n e^{-n^2 t}$$

$$\Rightarrow u(x, t) = \sum_{n=0}^{\infty} a_n e^{-n^2 t} \sin(nx)$$

$t=0$:

$$u(x, 0) = \sum_{n=1}^{\infty} a_n \sin(nx) = f(x) \quad / \cdot \sin(nx) / \int_0^{\pi} dx$$

$$a_n \int_0^{\pi} \sin^2(nx) dx = \int_0^{\pi} \sin(nx) dx$$

$$\vdots \quad \pi/2$$

$$a_n \frac{\pi}{2} = \frac{1}{n} \left(\cos\left(\frac{n\pi}{2}\right) - \cos(n\pi) \right)$$

$$\rightarrow u(x, t) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left(\cos \frac{n\pi}{2} - \cos n\pi \right) e^{-n^2 t} \sin(nx)$$